What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Acyanotic Congenital Heart Defects (Left–to–Right Shunts)

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Adapted from: Patent Ductus Arteriosus

Listen to this Brick
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Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of acyanotic congenital heart defects.

With a PDA, the pressure in the descending aorta is much greater than the pressure in the pulmonary artery, causing blood to flow continuously from the descending aorta into the pulmonary artery, creating a left-to-right shunt. This causes two problems:

- The left side of the heart cannot efficiently pump blood. Because some of the blood ejected from the LV reenters the pulmonary circulation, the heart must work harder to get the proper volume of blood into the systemic circulation. This leads to compensatory remodeling (hypertrophy) of the LV.
- Increased blood flowing into the pulmonary circulation from the RV and the PDA causes the left atrium (LA) and LV (which receive this blood) to dilate, and the pulmonary vasculature may thicken to accommodate larger blood volumes.

Complications:

Heart Failure. Chronic volume overload of the left side of the heart can lead to impaired functioning of the LA and LV.

Pulmonary Hypertension. Chronic volume overload of pulmonary circulation can lead to remodeling (thickening) of pulmonary vasculature, which increases resistance to blood flow and causes a compensatory elevation of pulmonary arterial pressures.

Eisenmenger Syndrome. The resistance of pulmonary circulation can become so great that it can lead to compensatory hypertrophy of the right side of the heart and eventual re s the PDA (blood flow from pulmonary arteries into the descending aorta). The development of new right-to-left shunt in the setting of CHD is known as Eisenmenger syndrome.

Such PDA patients may show **differential cyanosis** with only the lower extremities appearing blue. This occurs because in the right-to-left shunting, the deoxygenated blood will enter the aorta distal to where the arteries supplying the upper extremities exit the aorta (ie, left subclavian artery). So the upper extremities are supplied with normally oxygenated blood while blood supply to the lower extremities is reduced, causing lower extremity cyanosis (Figure 3).

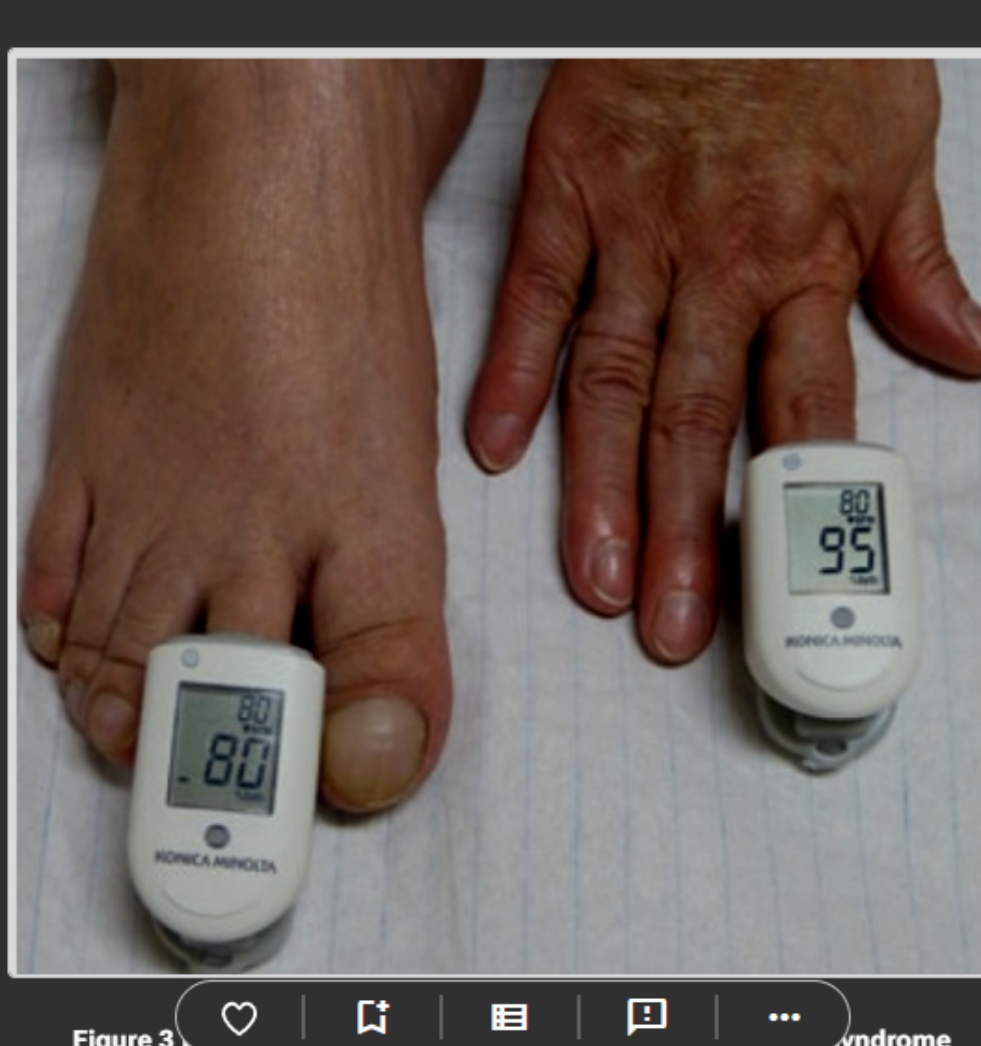
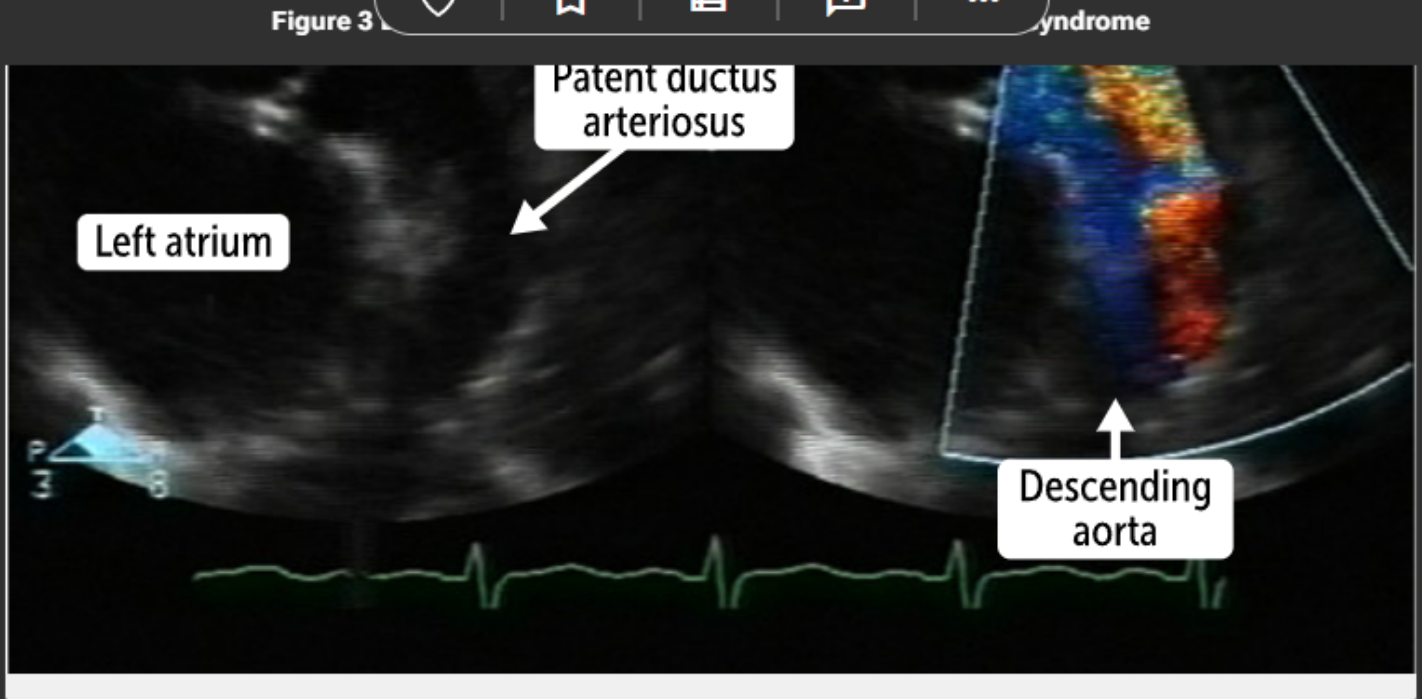


Figure 3 yndrome



QUIZ

Figure 4 Echocardiography in patent ductus arteriosus

Doppler can detect the shunting of blood through the PDA and can be used to assess disease severity. Doppler flow studies quantify the flow through the PDA.

Occasionally, cardiac catheterization is used to measure changes in the oxygen saturation that arise because of cardiac defects. In a heart with a PDA, cardiac catheterization will demonstrate that the blood oxygen saturation in the pulmonary artery is higher than the blood oxygen saturation in the RV, because oxygenated blood is being shunted from the aorta into the pulmonary artery. (This elevation is known as a "step-up.")

Management

Premature infants with PDA are initially managed medically with the use of

What Is Atrial Septal Defect? (LO#1)

An atrial septal defect (ASD) is a congenital heart defect in which an opening persists in the interatrial septum, the wall that normally separates oxygen-rich blood in the left atrium from the oxygen-poor blood in the right atrium. Because left atrial pressure is higher, blood moves through the defect from left to right, producing a **left-to-right shunt** (Figure 5).

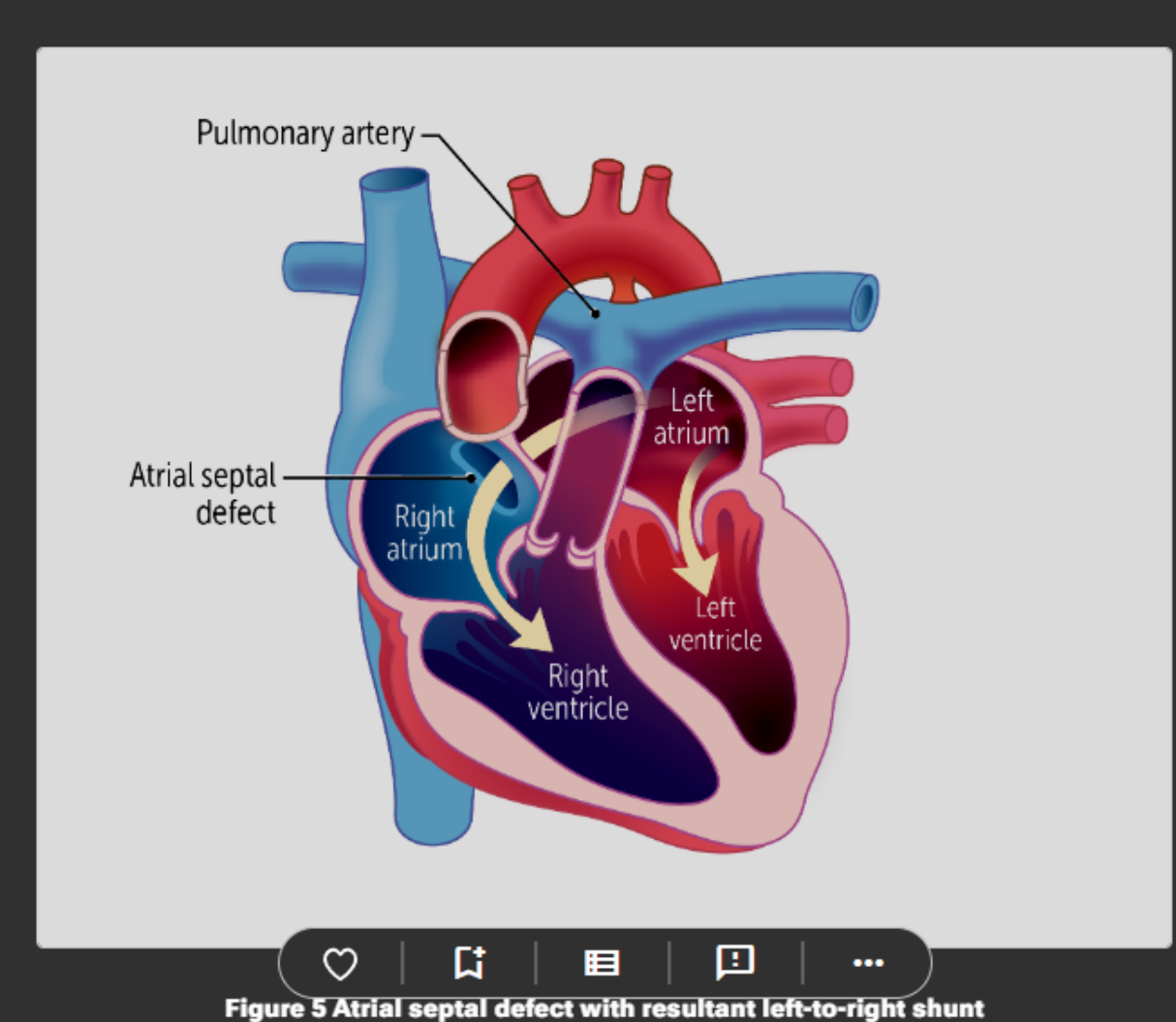


Figure 5 Atrial septal defect with resultant left-to-right shunt

Pathophysiology

An ASD is an abnormal communication between the RA and LA that persists after birth. To understand how this opening develops, it helps to recall the normal sequence of atrial septum formation during fetal life (Figure 6).

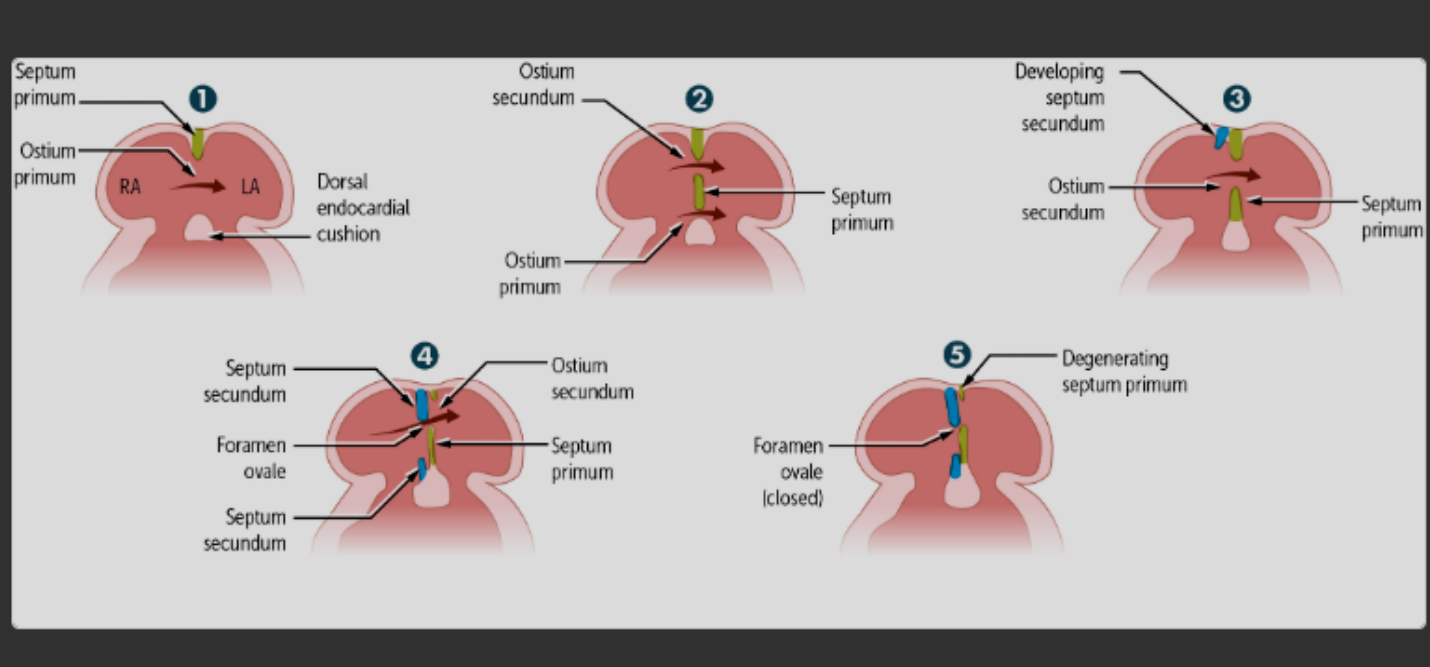


Figure 6 Atrial septum formation

The interatrial septum forms in two main steps:

1. **Septum primum formation** - A thin crescent of tissue grows downward from the roof of the primitive atrium toward the endocardial cushions, leaving a temporary opening called the **foramen primum**. As this gap closes, small perforations appear in the upper portion of septum primum that coalesce into the **foramen secundum**, allowing continued flow between the atria

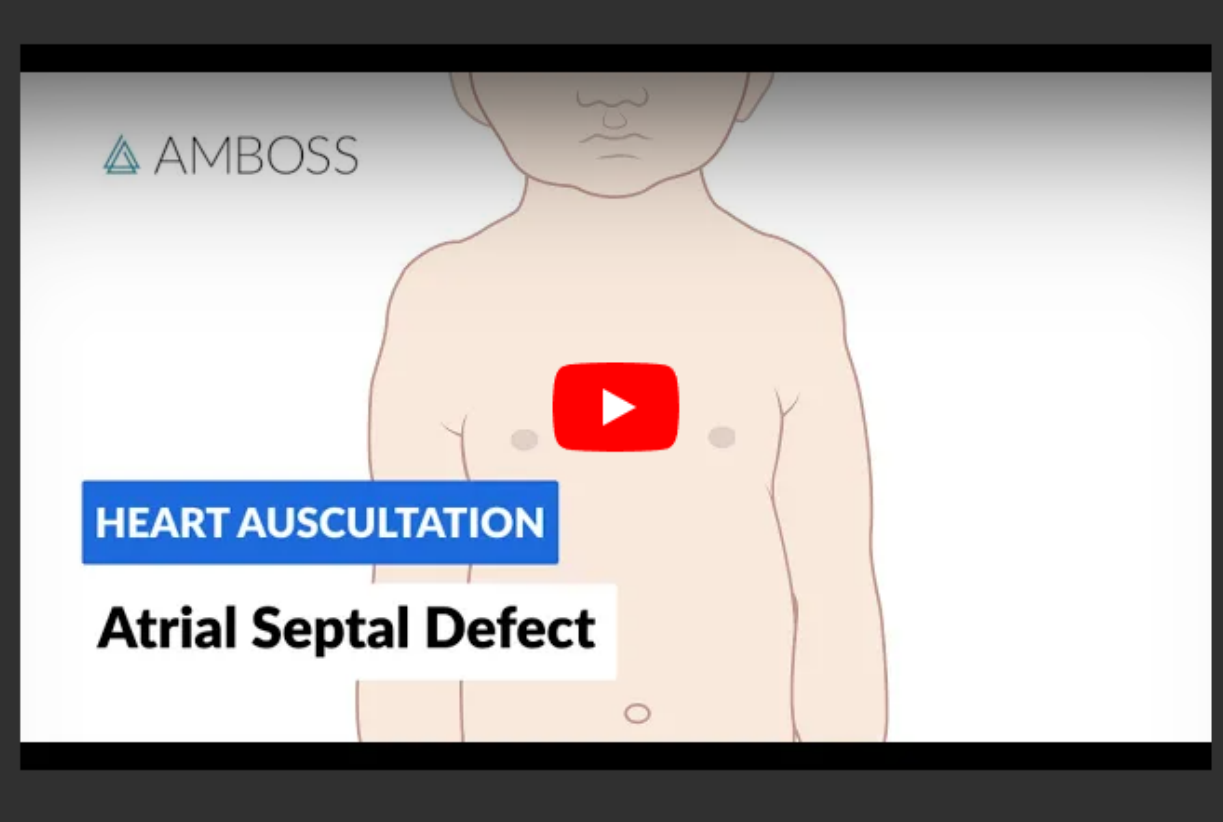
Ostium **Primum Defects.** (15% to 20% of ASDs) Occur lower in the atrial wall when the septum primum fails to fuse with the endocardial cushions. This results in persistence of the foramen primum. These defects are generally located in the ventral portion of the atrial septum.

Ostium Secundum Defects. (70% of ASDs) Result from excessive resorption of the septum primum or inadequate growth of septum secundum. They occur near the fossa ovalis in the central portion of the atrial septum. Recall that the fossa ovalis is a depression in the atrial septum that marks where the **foramen ovale** closed after birth. The foramen ovale is an opening with a flap that allows blood flow from the right atrium to the left atrium, with the flap closing off after atrial systole.

> **CLINICAL CORRELATION**
>
> A patent foramen ovale (PFO)—failure of the foramen ovale to close after birth—is not considered a true ASD because no atrial septal tissue is missing. PFOs occur in up to 25% of the normal population, and most people with a PFO are asymptomatic.

Sinus Venosus and Coronary Sinus Defects. Although these defects do not arise in the atrial septum per se, they are functionally similar to defects in the atrial septum because they allow blood from the LA to be shunted into the RA. Sinus venosus defect by a superior vena cava (SVC) or inferior vena cava (IVC) that abnormally enters the RA.

second left intercostal space radiating to the back (please see video). It increases with respiration.



Imaging

Once an ASD is suspected on physical exam, an **echocardiogram with Doppler** is used to confirm the diagnosis and assess the location, size, and direction of flow across the defect. Echocardiography also helps estimate the degree of chamber enlargement and pulmonary pressures.

In severe cases, cardiac catheterization can assess changes in oxygen saturation in different heart chambers. In a heart with an ASD, blood samples from RA, RV, and pulmonary artery would show a higher oxygen content than those in the SVC and

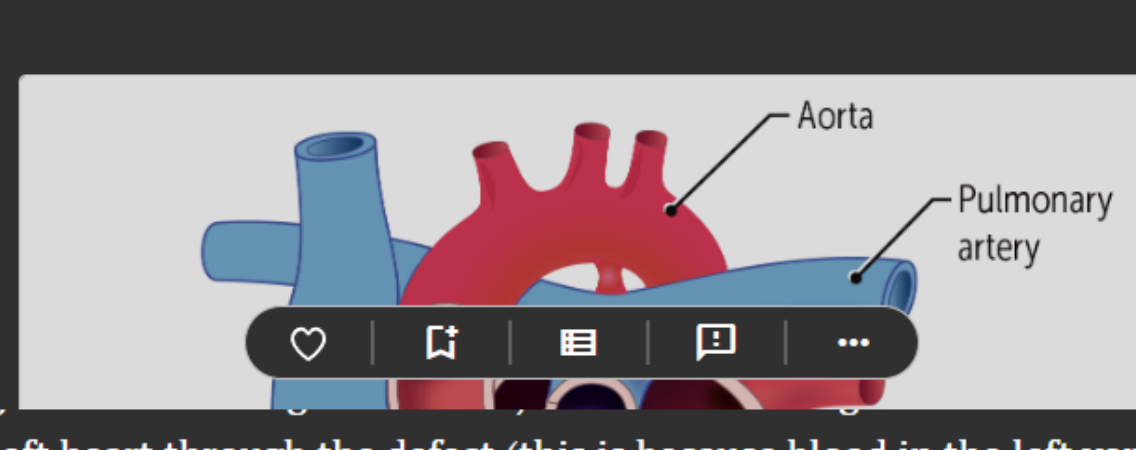
Management

For small ASDs, corrective intervention may be unnecessary, as spontaneous closure may occur in young patients. Defects less than 8 mm in diameter close on their own in nearly 80% by 15 months of age. The likelihood of spontaneous closure decreases with age, so persistent small ASDs that cause no significant shunting are usually monitored over time to ensure that continued non-intervention remains appropriate.

Larger defects—or small defects with a large amount of shunting—should be repaired even in the absence of symptoms. Interventions for repair include surgery or percutaneous closure. Percutaneous repair is now generally the procedure of choice because it is less invasive.

What Is Ventricular Septal Defect? (LO#1)

Ventricular septal defect (VSD) is the most commonly diagnosed congenital cardiac anomaly in infancy, occurring in 4 of every 1000 births. As the name suggests, a VSD is caused by an abnormal opening in the tissue of the interventricular septum, which normally separates the highly oxygenated blood of the left ventricle (LV) from the poorly oxygenated blood of the right ventricle (RV), as shown in Figure 9.



to the left heart through the defect (this is because blood in the left ventricle is generally under higher pressure than blood in the right ventricle, and blood moves down its pressure gradient). This means that poorly oxygenated blood (from the right heart) is not released into the circulation.

Pathophysiology

To understand the pathophysiology of a VSD, let's review the relevant steps in fetal cardiac development. The interventricular (IV) septum is composed of two parts: a membranous portion and a muscular portion (Figure 10).

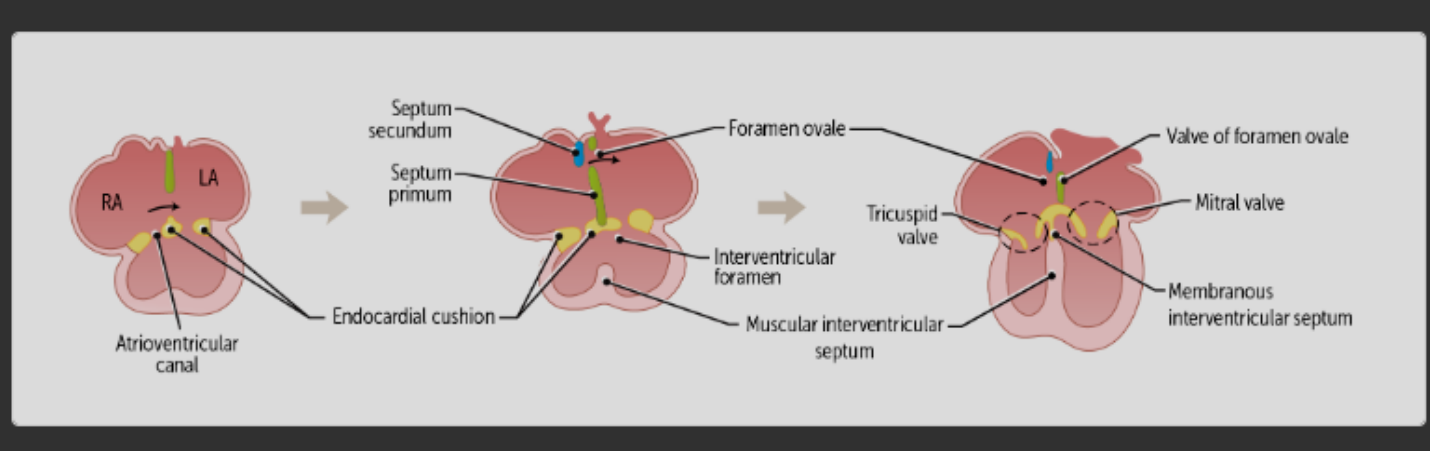


Figure 10 Interventricular septum formation

The membranous portion forms from the fusion of three structures: the endocardial cushions, the aorticopulmonary septum, and the muscular interventricular septum. The muscular portion arises from the floor of the single ventricle. If all proceeds normally during development, the muscular and membranous portions join together to complete the IV septum toward the end of the 7th week of gestation.

As you may have gue m does not always go according to plan, and errors in this process may result in a VSD. The

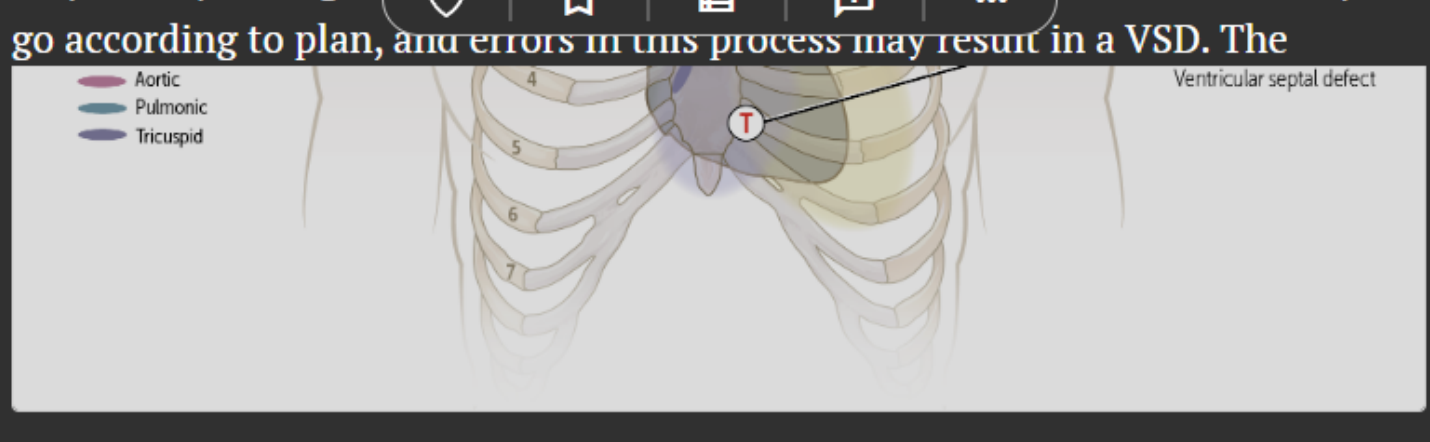


Figure 11 Auscultation of congenital murmurs

Smaller defects tend to have louder, higher-pitched murmurs because the turbulent flow across the defect is greater. In fact, turbulent flow can be so strong that sometimes a **palpable thrill** can be appreciated over the left sternal border. In contrast, larger defects have softer, lower-pitched murmurs.

The intensity of a VSD murmur may change over time. Although VSDs are present at birth, a murmur may not be heard immediately after delivery because the PVR is still quite high in the perinatal period. The elevated PVR leads to increases in RV pressures. The increases in RV pressures lower the pressure gradient between the LV and the RV, thus decreasing the left-to-right shunting across the VSD. As pulmonary resistance and right heart pressures decrease naturally after birth, the pressure gradient between the LV and RV increases, leading to greater shunting across the VSD. As a result, a murmur becomes audible. In contrast, murmur intensity may decrease over time if a moderate-to-large VSD is not corrected. This occurs because the ongoing remodeling of the pulmonary vasculature and RV leads to elevated RV pressures. Again, increased RV pressures cause a decrease in blood flow across the shunt.

Imaging

saturation in the RV and pulmonary artery.

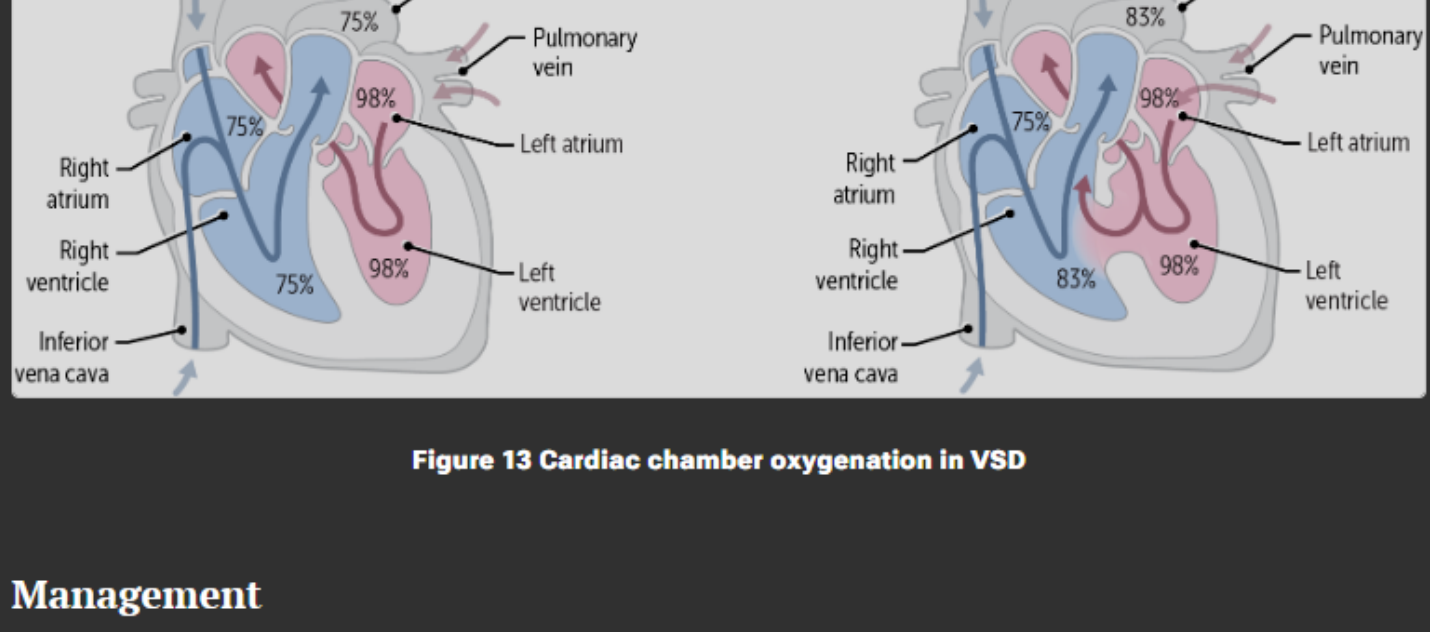


Figure 13 Cardiac chamber oxygenation in VSD

Management

Small to moderate VSDs do not require operative correction. In fact, about 50% of small to moderate VSDs undergo partial or complete spontaneous closure.

For larger defects in children with heart failure or pulmonary vascular disease, correction of the defect is recommended within the first few years of life to mitigate further pathologic remodeling of the heart and pulmonary vasculature. Catheter-based occlusion is generally preferred over surgery to correct these lesions.

> **CASE CONNECTION** BACK TO INTRODUCTION ↑
>
> Thinking back to LM, w will you explain
>
> diagnose PDA, but echocardiography is preferred because it is less costly and provides additional functional information that can be used to rate the severity. Further, avoiding ionizing radiation exposure would be especially desirable in a patient of this age. Electrocardiogram (D) and magnetic resonance imaging (E) can also detect findings associated with PDA, but neither is the preferred modality.

3. Which acyanotic CHD is characterized by a left-to-right shunt through an interventricular opening?

A. ○ Atrial Septal Defect
B. ● Ventricular Septal Defect
C. ○ Patent Ductus Arteriosus
D. ○ Tetralogy of Fallot

> **Hide Explanation**
>
> The correct answer is Ventricular Septal Defect (B). VSD involves a left-to-right shunt through an interventricular opening. Atrial Septal Defect (A) involves an interatrial opening. Patent Ductus Arteriosus (C) is a shunt via the ductus arteriosus. Tetralogy of Fallot (D) is a cyanotic defect, not acyanotic.

4. What is a major complication of untreated large Atrial Septal Defect (ASD)?

A. ● Pulmonary hypertension